Started on	Tuesday, 3 May 2022, 1:12 PM
State	Finished
Completed on	Tuesday, 3 May 2022, 2:33 PM
Time taken	1 hour 21 mins
Marks	6.00/7.00
Grade	**8.57** out of 10.00 (**86**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What component of the global compiler mechanism connects different object files into a complete program?

- a. The Preprocessor.
- b. The Linker. ✔
- c. The Compiler.

Your answer is correct.

The correct answer is:
The Linker.

Question **2**

Partially correct

Mark 0.50 out of 1.00

The simple counting program studied in last week's lecture is shown bellow.

```
#include <stdio.h>
#define STOP 0

/* Function: main
   Description: counts down from user input to STOP */
int main()
{
  // variable declarations
  int counter;  // an integer to hold count values
  int startPoint; // starting point for countdown
  printf("Enter a positive number: "); // prompt user for input
  scanf("%d", &startPoint);  // read into startPoint
  // count down and print count
  for (counter=startPoint; counter >= STOP; counter--)
    printf("%d\n", counter);
}
```

Among the following statements, select all which are **NOT TRUE**?

- [ ] a. The preprocessor directive *#include <stdio.h>* is required to use *printf* and *scanf* functions correctly.
- [x] b. The value returned by the main function can be used to assess the execution status when exiting the program. ✖
- [x] c. Two integer variables, counter and startPoint, are declared and initialised to 0. ✔ The two variables are declared, but not initialised to any value. They have a random value, until further modified in the program.
- [ ] d. Each element printed in the for loop is printed on a new line.
- [x] e. A macro defined by the directive *#define* cannot be modified by the program. ✖
- [ ] f. The startPoint variable is a pointer to an integer.

Your answer is partially correct.

You have selected too many options.
The correct answers are: Two integer variables, counter and startPoint, are declared and initialised to 0.,
The startPoint variable is a pointer to an integer.

Question **3**

Correct

Mark 1.00 out of 1.00

What is the output of the following C statement?

printf("4 multiplied by 8 in ascii is %c.\n", 4*8);

- (●) a. 4 multiplied by 8 in ascii is . ✔
- ( ) b. 4*8
- ( ) c. 4 multiplied by 8 in ascii is 4*8.
- ( ) d. 4 multiplied by 8 in ascii is 32.
- ( ) e. 4 multiplied by 8 in ascii is ♥☻.

Your answer is correct.

The correct answer is:
4 multiplied by 8 in ascii is .

Question **4**

Partially correct

Mark 0.50 out of 1.00

Consider the following C instruction:

scanf("%d %s", &first, second);

Among the following statements, select ALL which are **NOT TRUE**?

- [ ] a. The "&" symbol before "first" means that the user is inputting the address of the variable.
- [ ] b. The second input is a string value.
- [x] c. The statement reads two inputs and stores them in two variables: first, second. ✖
- [x] d. There must be exactly one space between the inputs when typed by the user. ✔ scanf allows unspecified white space between inputs.
- [ ] e. The first input is a decimal integer.

Your answer is partially correct.

You have correctly selected 1.
The correct answers are:
There must be exactly one space between the inputs when typed by the user.,

The "&" symbol before "first" means that the user is inputting the address of the variable.

Question **5**

Correct

Mark 1.00 out of 1.00

Consider the following C statements:

```
int var1 = 55;
int var2 = 110;
int var3 = 22;
int *pt1 = &var1;

pt1 = &var2;
*pt1 = var1;
var1 = var3;
var3 = var2;
pt1 = &var1;
*pt1 = var3;
```

What are the values of **var1**, **var2** and **var3** at the end of the statements?

- a. var1 = 110, var2 = 22 and var3 = 55
- b. var1 = 55, var2 = 55 and var3 = 55 ✔
- c. var1 = 55, var2 = 110 and var3 = 22
- d. var1 = 22, var2 = 55 and var3 = 110
- e. var1 = 110, var2 = 110 and var3 = 110

Your answer is correct.

The correct answers are:
var1 = 55, var2 = 110 and var3 = 22 ,

var1 = 55, var2 = 55 and var3 = 55

Question **6**

Correct

Mark 1.00 out of 1.00

Consider the following C program:

```
#include <stdio.h>

void mess(int *a, int b) {
    int t;
    t = *a;
    *a = b;
    b = t;
}

int main(void) {
    int x, y;
    x = 4;
    y = 5;
    mess(&x, y);
    printf("x = %d, y = %d\n", x, y);
}
```

What will be the output of the program?

- a. x = 4, y = 5
- b. x = 5, y = 5 ✔
- c. x = 4, y = 4
- d. x = 5, y = 4

Your answer is correct.

The correct answer is: x = 5, y = 5

Question **7**

Correct

Mark 1.00 out of 1.00

Consider the following C program:

```
#include <stdio.h>

void mess(int *a, int *b) {
    int t;
    t = *a;
    *a = *b;
    *b = t;
}

int main(void) {
    int x, y;
    x = 4;
    y = 5;
    mess(&x, &y);
    printf("x = %d, y = %d\n", x, y);
}
```

What will be the output of the program?

- a. x = 4, y = 5
- b. x = 4, y = 4
- c. x = 5, y = 4 ✔
- d. x = 5, y = 5

Your answer is correct.

The correct answer is:
x = 5, y = 4

◄ Practice MCQ 6 (for practice only)

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C Programming Practice ►